

Fecal carbohydrate-degrading bacteria are associated with reduced incidence of lower gastrointestinal GVHD

Christopher Fan,^{1,3} Tomo Hayase,^{1,4} Chia-Chi Chang,¹ Israel K. Glover,¹ Ivonne I. Flores,¹ Lauren K. McDaniel,¹ Miriam R. Ortega,¹ Christopher A. Sanchez,¹ Rawan K. El-Himri,¹ Alexandria N. Brown,¹ Jennifer L. Karmouch,¹ Mohamed A. Jamal,¹ Saira S. Ahmed,¹ Taylor M. Halsey,¹ Yimei Jin,¹ Wen-Bin Tsai,¹ Rishika Prasad,¹ Altai Enkhbayar,¹ Aqsa Mohammed,¹ Maren Schmiester,¹ Ashish Damania,¹ Nadim J. Ajami,¹ Jennifer A. Wargo,¹ Christine B. Peterson,⁵ Gabriela Rondon,⁶ Taha Al-Juhaishi,^{6,7} Amin M. Alousi,⁶ Jeffrey J. Molldrem,⁶ Richard E. Champlin,⁶ Elizabeth J. Shpall,⁶ Eric Martens,⁸ Cesar A. Arias,^{3,9,10} Robert R. Jenq,^{1,6,11,*} and Eiko Hayase^{1,4,*}

¹Department of Genomic Medicine, The University of Texas MD Anderson Cancer Center, Houston, TX; ²Division of Gastroenterology, Department of Medicine, Houston Methodist Academic Institute, Houston Methodist Hospital, Houston, TX; ³Department of Medicine, Weill Cornell Medical College, New York, NY; ⁴Section of Hematology and Oncology, Department of Medicine, Baylor College of Medicine, Houston, TX; ⁵Department of Biostatistics and ⁶Department of Stem Cell Transplantation and Cellular Therapy, The University of Texas MD Anderson Cancer Center, Houston, TX; ⁷Section of Hematology and Medical Oncology, Department of Medicine, Stephenson Cancer Center, University of Oklahoma Health Sciences Center, Oklahoma City, OK; ⁸Department of Microbiology and Immunology, University of Michigan Medical School, Ann Arbor, MI; ⁹Division of Infectious Diseases, Department of Medicine, Houston Methodist Academic Institute, Houston Methodist Hospital, Houston, TX; ¹⁰Center for Infectious Diseases, Houston Methodist Research Institute, Houston, TX; and ¹¹Department of Hematology and Hematopoietic Cell Transplantation, City of Hope National Medical Center, Duarte, CA

Key Points

- Carbohydrate-degrading function of the gastrointestinal microbiota is important in mediating risk of LGI-GVHD.

Lower gastrointestinal graft-versus-host disease (LGI-GVHD) carries morbidity and mortality for patients undergoing allogeneic hematopoietic stem cell transplantation (allo-HSCT), with critical contributions from the intestinal microbiome. In a retrospective cohort of metagenomic sequencing of stool from patients with allo-HSCT (N = 90), we found that a reduction in specific *Parabacteroides* and *Bacteroides* species around the time of engraftment contributes to LGI-GVHD risk. Given the known diverse carbohydrate-degrading functionality of these bacteria, we investigated gene abundances for carbohydrate-active enzymes (CAZymes) and found that *Parabacteroides merdae*, *P distasonis*, and *Bacteroides ovatus* abundances were significantly correlated with CAZymes in patients who did not develop LGI-GVHD compared with those who did. The specific gene abundances of xylosidase, which contribute to the degradation of xylose-containing polysaccharides, were significantly associated with a reduced risk of LGI-GVHD. All these findings show the importance of the carbohydrate-degrading functionality of putative beneficial bacteria in mediating risk of LGI-GVHD.

Introduction

Allogeneic hematopoietic stem cell transplantation (allo-HSCT) is an established curative therapy for the treatment of hematologic malignancies. However, posttherapy complications such as graft-versus-host disease (GVHD) can complicate the posttransplant recovery process and increase morbidity and

Submitted 10 April 2025; accepted 18 November 2025; prepublished online on *Blood Advances* First Edition 22 January 2026; final version published online 13 March 2026. <https://doi.org/10.1182/bloodadvances.2025016780>.

*R.R.J. and E.H. contributed equally to this study.

Whole-genome sequencing data of patient fecal samples have been deposited in the Sequence Read Archive (accession number PRJNA1094987). No specific code was developed for analysis.

All data generated from this study are available from the corresponding author, Eiko Hayase (eiko.hayase@bcm.edu), on request.

The full-text version of this article contains a data supplement.

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mortality associated with the treatment.¹ Certain microbiome and functional changes are known to be associated with an increased incidence and severity of lower gastrointestinal GVHD (LGI-GVHD), with specific contributions from exposure to broad-spectrum antibiotics such as carbapenems and activity from carbohydrate-degrading bacteria.^{2,3} Short-chain fatty acids (SCFAs), which are byproducts of bacteria-mediated fiber degradation, have also been shown to suppress the incidence of LGI-GVHD.^{4,5}

Bacteroides spp. have a diverse repertoire of carbohydrate-degrading enzymes with gene groups predicted to code for carbohydrate-active enzyme (CAZyme) families, designated polysaccharide utilization loci (PUL), which coordinate complex carbohydrate degradation. Our group previously reported that increased the mucus-degrading function of *Bacteroides thetaio-tomicron* induced severe LGI-GVHD, whereas the degradation of xylose-containing polysaccharide by *Bacteroides ovatus* contributed to suppressing the severity of LGI-GVHD in a pre-clinical model.^{2,3} These data suggested that complex carbohydrate degradation by intestinal microbes is important for the regulation of alloreactive immune cells in allo-HSCT. However, the role of bacteria-mediated complex carbohydrate degradation in regulating LGI-GVHD in patients receiving allo-HSCT is unclear. In this study, we aimed to identify taxonomical and functional factors that are associated before the development of LGI-GVHD, with a focus on how alterations in CAZyme gene abundances, PUL, and specific enzyme functions contribute to suppressing LGI-GVHD.

Materials and methods

Retrospective cross-sectional study design and clinical cohort

A total of 90 patients who underwent allo-HSCT for acute myeloid leukemia/myelodysplastic syndromes from May 2017 to February 2020 at the MD Anderson Cancer Center provided stool samples for our biorepository. These patient stool samples were analyzed retrospectively, during which stool samples were collected after HSCT, before the development of acute LGI-GVHD. Acute LGI-GVHD was diagnosed by clinical and/or pathological findings and graded according to standard criteria.⁶ We determined systemic glucocorticoid steroid response as previously reported.⁷ Briefly, no response was defined as a lack of response after at least 3 days of high-dose systemic glucocorticoid therapy, a lack of improvement after 7 days, or treatment failure during steroid tapering or an inability to taper the dose to <0.5 mg/kg per day of methylprednisolone. We did not select a control cohort matched for clinical parameters such as comorbidity index, involvement of other GVHD target organs, or chronic GVHD status in this study.

Antibiotic exposure

Patient antibiotic exposure data were gathered from the electronic medical records from day -7 of HSCT to day 14 post-HSCT. All formulations of the antibiotics, excluding topically administered antibiotics, were included in the analysis. Patients with antibiotic exposure >20% of the time from day -7 of HSCT to day 14 post-HSCT were deemed to be a positive antibiotic exposure.

Whole-genome sequencing of patient fecal samples

Genomic DNA was isolated from patient fecal samples and purified using a Qiagen Genomic-tip 20/G column, according to the manufacturer's instructions. For short-read Illumina sequencing, libraries were constructed with a Nextera DNA Flex Library Prep Kit (Illumina), according to the manufacturer's protocol. All libraries were quantified with a TapeStation and pooled in equal molar ratios. The final libraries were sequenced with a NovaSeq 6000 platform (Illumina) to produce 2 × 150 bp paired-end reads, resulting in ~5 Gb per sample. In sequencing analysis, sequence reads were filtered by their quality using the VSEARCH 2.17.1. The abundance of taxa was profiled using HUMAnN3. HUMAnN3 incorporates the metaphlan pipeline, which generates raw counts, utilizing the flag `-metaphlan-options "-t rel_ab_w_read_stats,"` which provides relative abundances and total counts. Taxonomic profiling using HUMAnN3 indicated that ~90% of the total reads were derived from bacteria.

Microbiome data analysis

Relative species level abundances were calculated after normalizing raw count data by the total counts per sample. α -Diversity measures were calculated using the Shannon index, and β -diversity measures were calculated through the Bray-Curtis distances using the vegan package in R software version 4.3.1. To conduct multidimensional scaling, `cmdscale()` was used in R. Principal coordinates 1 and 2 and their respective percent variance explained are shown on the x- and y-axes. Groups were compared via PERMANOVA using `adonis2` in the vegan R package. Differential abundance profiles were analyzed using the MaAsLin2 package in R, with q values reported corrected for false discovery rate (FDR).⁸ Thresholds of minimum prevalence of 0.25 and minimum abundance of 0 were set.

CAZyme alignment

The qualities of sequence data were filtered and truncated on their quality by VSEARCH using default settings. The sequences of interest were further identified using DIAMOND software (version 0.9.24)⁹ to align against the Carbohydrate-Active Enzymes database (<http://www.cazy.org/>), and we selected the best match of alignments using `-max-target-seqs 1` option in DIAMOND. Features with percent identity <80% were excluded. Aligned reads were filtered by identifying the organisms of accession numbers and only including those that aligned to bacteria. To determine differentially abundant CAZymes, we compared sequences of those that went on to develop LGI-GVHD vs those that had no LGI-GVHD using the DESeq2 package in R via RStudio (version 1.2.1335).¹⁰ Volcano plots were generated using the EnhancedVolcano package in R (<https://github.com/kevinblighe/EnhancedVolcano>). *P* values were adjusted by the Benjamini-Hochberg correction.¹¹ An adjusted *P* value of <.05 was considered statistically significant. Putative substrates were mapped using results from dbCAN-seq.¹²

Network analysis

Pearson correlation between logit-transformed CAZyme gene family and species-level taxa data was conducted. *P* values were FDR corrected using the Bonferroni correction method, and significance was set at 0.05. Network visualization was performed using Cytoscape version 3.10.1.

Table 1. Clinical characteristics of patients with acute myeloid leukemia/myelodysplastic syndrome who underwent allo-HSCT

	Overall (N = 90)	Non-LGI-GVHD (n = 74)	LGI-GVHD (n = 16)	P value
Age at transplant, median (IQR), y	60 (46-66)	60 (47-67)	58 (43-65)	.3
Male, n (%)	52 (57.8)	43 (58.1)	9 (56.2)	> .9
Donor type, n (%)				.2
Matched sibling	21 (23.3)	16 (21.6)	5 (31.2)	
Haploidentical	17 (18.9)	14 (18.9)	3 (18.8)	
Mismatched unrelated donor	4 (4.4)	2 (2.7)	2 (12.5)	
Matched unrelated donor	48 (53.3)	42 (56.8)	6 (37.5)	
Cell source, n (%)				.4
Peripheral blood	64 (71.1)	53 (71.6)	11 (68.8)	
Bone marrow	24 (26.7)	20 (27.0)	4 (25.0)	
Cord blood	2 (2.2)	1 (1.4)	1 (6.2)	
Nonmyeloablative, n (%)	23 (25.6)	18 (24.3)	5 (31.2)	.7
HCT comorbidity index, n (%)				
Low risk (0)	15 (16.7)	14 (18.9)	1 (6.2)	
Intermediate risk (1 or 2)	28 (31.1)	23 (31.1)	5 (31.2)	
High risk (≥ 3)	47 (52.2)	37 (50.0)	10 (62.6)	.4
GVHD prophylaxis, n (%)				.2
PTCy based	73 (81.1)	62 (83.8)	11 (68.8)	
Calcineurin inhibitor based	17 (18.9)	12 (16.2)	5 (31.2)	
GVHD organ involvement, n (%)				
Upper GI	27 (30.0)	20 (27.0)	7 (43.8)	.19
Liver	3 (3.3)	3 (4.1)	0 (0)	.41
Skin	27 (30.0)	21 (28.4)	6 (37.5)	.47
Maximum acute GVHD grade				
Grade 1	24 (26.7)	21 (28.4)	3 (18.8)	
Grade 2	9 (10)	5 (6.8)	4 (25.0)	
Grade 3	11 (12.2)	7 (9.5)	4 (25.0)	
Grade 4	6 (6.7)	1 (1.5)	5 (31.2)	
Unknown	1 (1.1)	1 (1.5)	0 (0)	
None	39 (43.3)	39 (52.7)	0 (0)	<.001
Time of onset LGI-GVHD, median (IQR), d		N/A	47 (30, 82)	
LGI-GVHD stages, n (%)				
Stages 1-2	N/A	N/A	7 (43.8)	
Stages 3-4	N/A	N/A	9 (56.2)	N/A
Steroid-responsive LGI-GVHD, n (%)	N/A	N/A	10 (62.5)	
Chronic GVHD, n (%)				
No	73 (81.1)	60 (81.1)	13 (81.3)	
Yes	11 (12.2)	10 (13.5)	1 (6.2)	
Not eligible	6 (6.7)	4 (5.4)	2 (12.5)	.46

IQR, interquartile range; N/A, not applicable; PTCy, posttransplant cyclophosphamide.

PUL gene analysis

Mapped PUL genes for the 3 target species (*B ovatus* ATCC_8483 new_assembly, *Parabacteroides merdae* CL03T12C32, and *Parabacteroides distasonis* F1-28) were stratified based on LGI-GVHD status. Differentially abundant genes were identified using the DESeq2 package in R.¹⁰ DESeq2 *P* values were FDR corrected,

and significance was set at 0.05. Results were plotted using the EnhancedVolcano package in R.¹³

Statistical analysis

Data were checked for normality and similar variances between groups, and Student *t* tests were used when appropriate. The

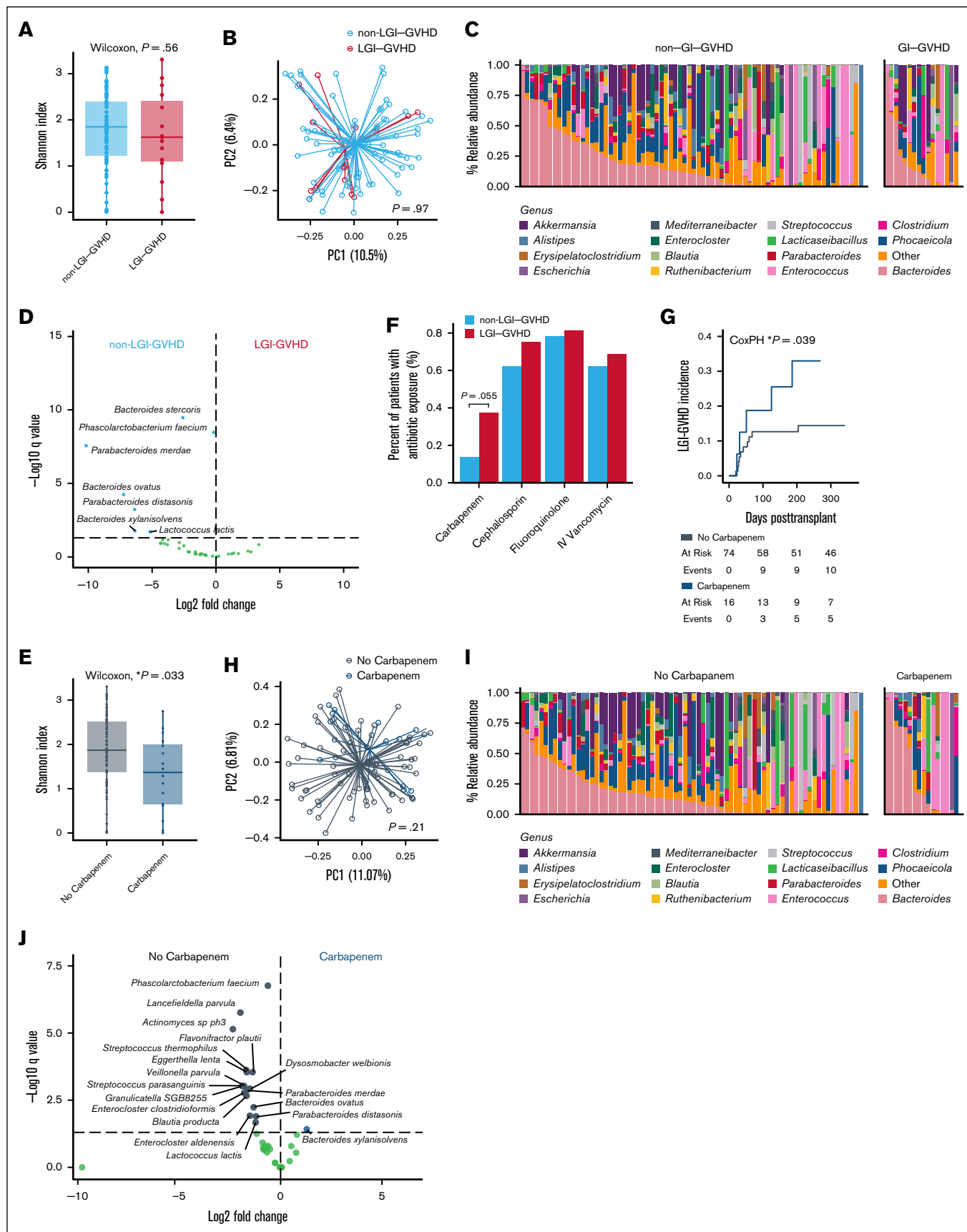


Figure 1.

Mann-Whitney U tests were used to compare nonnormally distributed data. Kaplan-Meier curves were used to show the incidence of acute LGI-GVHD, and the Cox proportional hazards model with Firth penalized likelihood was applied to test for association with the hazard using the `coxphf` package in R. For LGI-GVHD outcome in the model, patients with relapse of disease or death before the development of LGI-GVHD were censored. For clinical data analysis, nonrepeated analysis of variance was used to compare continuous variables, whereas the χ^2 or Fisher exact tests were used to analyze the frequency distribution between categorical variables. Analyses were performed using R and Prism version 9.0 (GraphPad Software). P values <0.05 were considered statistically significant.

Signed informed consent was provided by all study participants, and this study was approved by The University of Texas MD Anderson's institutional review board (IRB; collection IRB: LAB99-062, sequencing IRB: PA17-035).

Results

Fecal whole-genome sequencing identifies differentially abundant species in patients with non-LGI-GVHD vs LGI-GVHD

We first aimed to investigate the role of the fecal microbiome in identifying allo-HSCT patients at risk for developing acute LGI-GVHD using whole-genome sequencing. We retrospectively analyzed the stool samples collected around the time of engraftment (day 13 interquartile range [IQR], day 12-14) from 90 patients with acute myeloid leukemia/myelodysplastic syndrome who underwent allo-HSCT at MD Anderson Cancer Center from May 2017 to February 2020 (Table 1). This cohort included 74 patients with non-LGI-GVHD and 16 patients with LGI-GVHD. There were no significant differences in the allo-HSCT-related clinical parameters between patients with and without LGI-GVHD. However, patients with LGI-GVHD had significantly higher grades of acute GVHD than those without LGI-GVHD, reflecting the involvement of the lower gastrointestinal tract (Table 1). In an examination of the microbiome of the stool samples using whole-genome sequencing, α -diversity measures using the Shannon index did not differ between patients with LGI-GVHD and patients with non-LGI-GVHD (Figure 1A). β -Diversity measures at the species level using the Bray-Curtis distances and compared via PERMANOVA also demonstrated no significant difference between patients with LGI-GVHD and patients with non-LGI-GVHD (Figure 1B). The species level differential abundance analysis using MaAsLin2 showed specific taxa that favored the

absence of LGI-GVHD, including *B stercoris*, *Phascolarctobacterium faecium*, *P merdae*, *B ovatus*, *P distasonis*, *B xylanisolvens*, and *Lactococcus lactis* (Figure 1C-D). No specific species taxa favored the development of LGI-GVHD.

Given that broad-spectrum antibiotic exposure has been known to alter the microbiome composition and is associated with an increased risk of LGI-GVHD incidence, specifically related to carbapenems,^{3,14,15} we asked whether antibiotic exposure between days -7 and 14 from transplant was associated with an incidence of LGI-GVHD. Most antibiotics used were either carbapenems, cephalosporins, fluoroquinolones, or IV vancomycin. Prior studies have shown that meropenem decreases microbiome diversity in patients undergoing allo-HSCT.¹⁵ To avoid including the limited effect of short-term antibiotic exposure, we first determined the threshold at which the duration of antibiotic exposure affects α -diversity on day 14. We tested carbapenem exposure thresholds of 0%, 10%, and 20% relative to the total analyzed period from day -7 to day +14. Notably, a 0% and 10% carbapenem exposure threshold did not significantly affect α -diversity ($P = .1$ and $P = .16$, respectively), whereas a 20% carbapenem exposure threshold was the first to show a significant decrease (Figure 1E). In contrast, no exposure threshold for cephalosporins was associated with significant differences in α -diversity. Based on these findings, we defined patients with $>20\%$ antibiotic exposure from day -7 to day +14 relative to HSCT as having significant antibiotic exposure. Although not statistically significant, carbapenem exposure trended with the incidence of LGI-GVHD ($P = .055$), and the difference in the percent of patients with specific antibiotic exposure when comparing patients with LGI-GVHD vs patients non-LGI-GVHD was most pronounced in patients receiving carbapenems (Figure 1F). Using time to event analysis, we found that carbapenem exposure between days -7 and 14 from transplant was significantly associated with an increased incidence of LGI-GVHD, consistent with the prior studies^{3,14,16} (Figure 1G). We then investigated the impact of the use of carbapenem on the composition of the microbiome in patients undergoing allo-HSCT. As described earlier, carbapenem-exposed patients had a lower microbiome α -diversity (Figure 1G), whereas there was no difference in β -diversity between carbapenem and non-carbapenem-exposed patients (Figure 1H). Differential abundance analysis at the species level showed multiple bacterial species that favored non-carbapenem exposure, including *Ph faecium*, *P merdae*, *B ovatus*, *P distasonis*, and *L lactis*. Notably, most bacteria that favored the absence of LGI-GVHD were also shown to favor non-carbapenem exposure (Figure 1I-J), suggesting that the use of broad-spectrum antibiotics such as carbapenem was a key covariate associated with microbiome alterations that are in turn associated with poor clinical outcomes.

Figure 1. Carbapenem exposure and species-level taxa differences precede LGI-GVHD development. (A-J) The intestinal microbiome analyzed by whole-genome sequencing in patient stool samples collected around engraftment. (A) Species-level α -diversity measured by the Shannon index stratified by LGI-GVHD status. (B) PCoA (principal coordinates analysis) plot showing β -diversity measures stratified by LGI-GVHD status. P value was analyzed with PERMANOVA. (C) Stacked bar graph of genus-level relative abundance stratified by LGI-GVHD status. (D) Volcano plot showing species-level differential abundance analysis by LGI-GVHD status, showing $-\log_{10}$ q value vs $\log_2$ fold-change using MaAsLin2. (E) Species-level α -diversity measured by the Shannon index stratified by carbapenem exposure. (F) Bar graph showing antibiotic exposure from day -7 to day +14 relative to transplant stratified by LGI-GVHD. Patients with antibiotic exposure $>20\%$ of the time from day -7 of HSCT to day 14 post-HSCT were deemed to have a positive antibiotic exposure. P value demonstrates χ^2 analysis. (G) Kaplan-Meier curve of incidence of LGI-GVHD stratified by carbapenem exposure. (H) PCoA plot showing β -diversity measures stratified by carbapenem exposure. P value was analyzed with PERMANOVA. (I) Stacked bar graph of genus-level relative abundance stratified by carbapenem exposure. (J) Volcano plot showing species-level differential abundance analysis by carbapenem exposure status, showing $-\log_{10}$ q value vs $\log_2$ fold-change using MaAsLin2. CoxPH, cox proportional hazards model; PC1, principal coordinate 1.

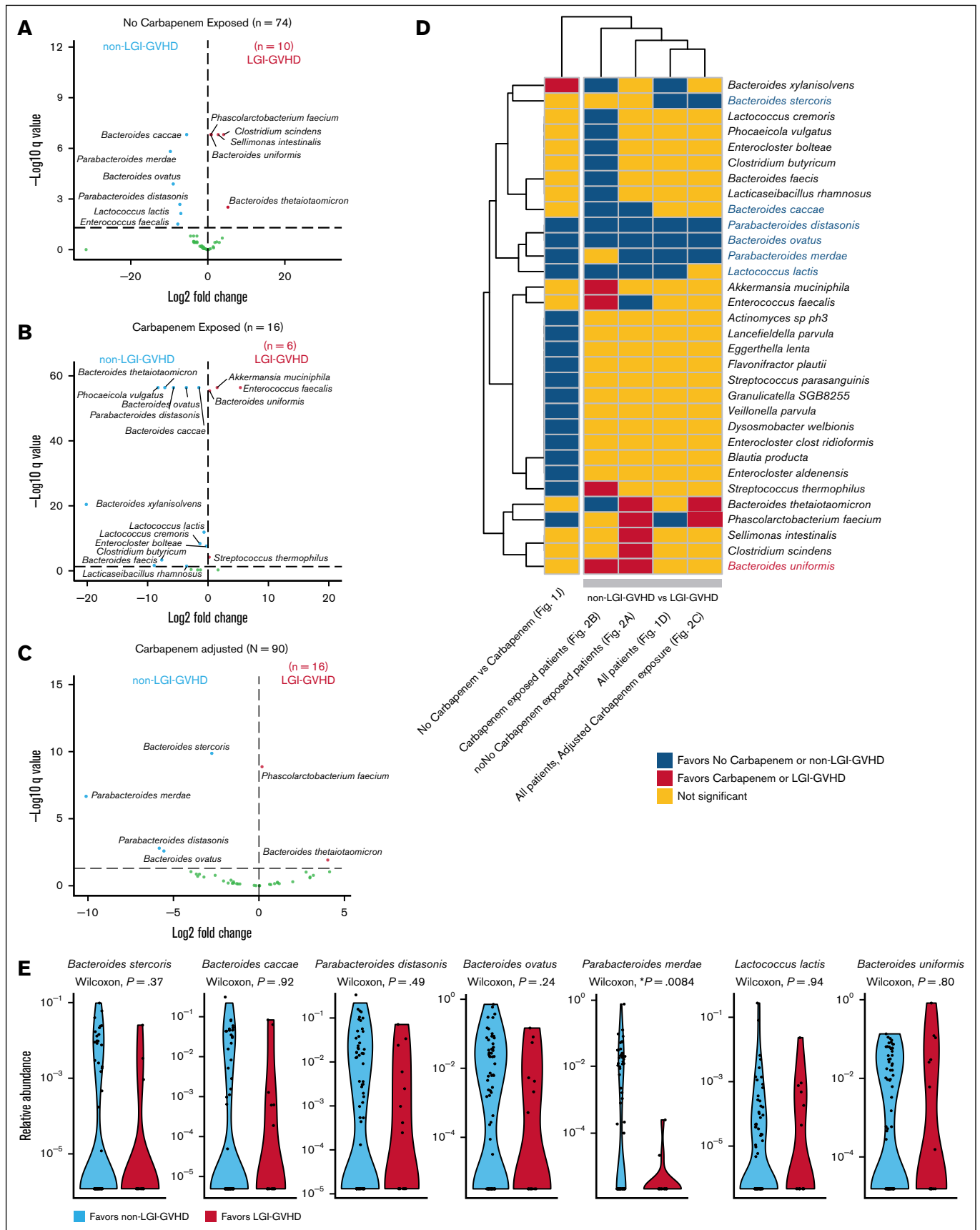


Figure 2.

Six bacterial species may modulate the risk of LGI-GVHD

To investigate microbiome composition associations with LGI-GVHD independent of carbapenem exposure, we investigated species-level differential abundance among non-carbapenem-exposed ($n = 74$) and carbapenem-exposed ($n = 16$) patient subsets. Among the patients not exposed to carbapenem, 6 species were shown to favor the absence of LGI-GVHD, including 2 *Bacteroides* species *B. ovatus* and *B. caccae* (Figure 2A). Five bacterial species were shown to favor LGI-GVHD, including *Ph. faecium*, *Sellimonas intestinalis*, *B. thetaiotaomicron*, *B. uniformis*, and *B. stercoris*. Among the patients exposed to carbapenem, 12 taxa were associated with a reduced risk of LGI-GVHD; notably, 4 taxa were from the genus *Bacteroides* including *B. thetaiotaomicron*, *B. ovatus*, *B. xylanisolvens*, and *B. faecis* (Figure 2B). *Akkermansia muciniphila* and *Enterococcus faecalis* were associated with an increased risk of LGI-GVHD. Given that there was a small cohort of carbapenem-exposed patients in this study, we also evaluated species-level differential abundance between LGI-GVHD and non-LGI-GVHD in all patients in our cohort ($N = 90$), while adjusting for carbapenem exposure using MaAsLin2 (Figure 2C). Four taxa were shown to favor the absence of LGI-GVHD, including *B. stercoris*, *P. merdae*, *P. distasonis*, and *B. ovatus*. Two taxa were shown to be associated with an increased risk of LGI-GVHD, including *Ph. faecium* and *B. thetaiotaomicron*.

Collectively, we identified 6 bacterial species that were favored in patients who did not develop LGI-GVHD and were not exposed to carbapenems, and these included *P. merdae*, *B. ovatus*, *P. distasonis*, *B. stercoris*, *B. caccae*, and *L. lactis* (Figure 2D). Although carbapenem exposure contributes to a decrease in these beneficial bacteria (Figure 1J), these bacteria favor the absence of LGI-GVHD even when analyzing only patients not exposed to carbapenems or after adjusting for carbapenem exposure, suggesting that other factors in addition to carbapenem exposure modulate the relative abundances of these bacteria and downstream LGI-GVHD risk. In general, the higher abundance of the genera *Bacteroides* and *Parabacteroides* has been known to contribute to food- or host-derived polysaccharide degradation in the colon.^{17,18} Recently, our group reported the role of carbohydrate metabolism by *B. ovatus* in reducing mucus utilization by mucolytic bacteria including *B. thetaiotaomicron* and *A. muciniphila* and suppressing GVHD-related mortality in a preclinical model.² We hypothesized that these bacteria suppress the incidence of LGI-GVHD through their carbohydrate-degrading function in the colon.

Carbohydrate degradation by *B. ovatus*, *P. distasonis*, and *P. merdae* may contribute to a reduced risk of LGI-GVHD

We then sought to investigate whether the genes of CAZymes were associated with the incidence of LGI-GVHD. α - and

β -diversity measures of CAZymes, based on the Shannon diversity and Bray-Curtis distances, respectively, did not differ between patients with LGI-GVHD and patients with non-LGI-GVHD (Figure 3A-B). We investigated differentially abundant CAZyme genes between patients who developed LGI-GVHD and patients with non-LGI-GVHD and identified 657 CAZyme genes, representing 163 unique CAZyme gene families, which were more abundant in patients with non-LGI-GVHD (Figure 3C). These data suggested that specific bacterial carbohydrate degradation functionalities are associated with a reduced incidence of LGI-GVHD. We examined the functional annotations of CAZyme families and their enzyme functions and identified multiple CAZyme families that favored non-LGI-GVHD, including xylosidase and glucosidase enzymatic activity (Figure 3D). Importantly, putative substrates of CAZyme families that favored non-LGI-GVHD utilize polysaccharides derived from plants and host (Figure 3E). Host-derived polysaccharide substrates include host glycans, mucin, and human milk oligosaccharides (supplemental Table 1). These data indicated that microbes capable of degrading of specific plant- and host-derived polysaccharides might contribute to a reduced risk of LGI-GVHD.

We then investigated the potential relationship between CAZyme gene families that favored non-LGI-GVHD and the abundance of specific species of *Bacteroides* and *Parabacteroides* that were generally favored and consistently present in patients with non-LGI-GVHD or not exposed to carbapenems (Figure 3F). We performed Pearson correlation analysis on logit-transformed CAZyme gene families and species-level taxa abundances (Figure 3F). Based on species that were shown to consistently favor non-LGI-GVHD and the relation of CAZymes and taxa abundances, we chose to focus on 3 specific putative beneficial bacterial species moving forward: *P. merdae*, *B. ovatus*, and *P. distasonis*. All 3 bacterial species were positively correlated with many CAZyme gene families that favored non-LGI-GVHD, suggesting that they actively degrade food- or host-derived carbohydrates via their carbohydrate-degrading functionalities in the colon. Interestingly, compared with *P. merdae*, *B. ovatus* and *P. distasonis* were more strongly correlated with CAZyme gene families (Figure 3G). This was redemonstrated by the number of CAZyme gene families correlated with each bacterial species, with *B. ovatus* and *P. distasonis* showing a higher number of correlated gene families than *P. merdae* (Figure 3G).

To characterize the function of these bacteria, we identified PUL from the 3 putative beneficial bacteria (*B. ovatus*, *P. distasonis*, and *P. merdae*). To perform this analysis, we used available PUL annotations from the genomes of strains representative of each of these species. Specifically, we chose *B. ovatus* ATCC_8483 (new_assembly), *P. merdae* CL03T12C32, and *P. distasonis* F1-28. We annotated sequencing reads using PUL gene data of these representative strains of the 3 beneficial bacteria in the PUL DataBase (PULDB). We then investigated differentially abundant

Figure 2. Carbapenem exposure affects the relative abundance of specific beneficial bacteria. (A) Volcano plot showing species-level differential abundance analysis by LGI-GVHD status among non-carbapenem-exposed patients, showing $-\log_{10}$ q value vs $\log_2$ fold-change using MaAsLin2. (B) Volcano plot showing species-level differential abundance analysis by LGI-GVHD status among carbapenem-exposed patients, showing $-\log_{10}$ q value vs $\log_2$ fold-change using MaAsLin2. (C) Volcano plot showing species-level differential abundance analysis by LGI-GVHD status and adjusted by carbapenem exposure, showing $-\log_{10}$ q value vs $\log_2$ fold-change using MaAsLin2. (D) Heat map showing the differentially abundant taxa favoring non-LGI-GVHD and LGI-GVHD across multiple comparisons. Red and blue indicate favoring LGI-GVHD and non-LGI-GVHD, respectively. (E) Violin plots showing differences in relative abundances when comparing non-LGI-GVHD and LGI-GVHD. P values reflect LGI-GVHD vs non-LGI-GVHD from Mann-Whitney U test.

PUL genes among these 3 strains and found that multiple PUL genes that belong to *B ovatus* PUL49, PUL94, and PUL42 favored the absence of LGI-GVHD (Figure 3H). For *P distasonis*, only a single PUL gene favored LGI-GVHD (Figure 3H). For *P merdae*, multiple PUL genes favored the absence of LGI-GVHD (Figure 3H) (supplemental Table 2).

CAZymes that harbor xylosidase function stratify LGI-GVHD risk

Our group previously reported that xylose supplementation or the introduction of *B ovatus*, which degrades xylose-containing polysaccharides into xylose in the colon, could each contribute to the reduced severity of meropenem-mediated LGI-GVHD in a pre-clinical model.^{2,3} We were then interested in evaluating whether the degradation of xylose-containing polysaccharides via xylanase and xylosidase gene abundances was associated with LGI-GVHD status in the clinical setting. We identified enzyme class numbers that mapped to xylanase and xylosidase functions. Specific enzyme classes of xylanase and xylosidase gene abundances did not statistically differ when stratified by LGI-GVHD status (Figure 4A). However, xylanase (and not xylosidase) gene abundances differed when comparing patients exposed to carbapenem vs not exposed to carbapenem (Figure 4B). Using cutoffs for enzyme class (EC) abundances determined from receiver operator curves, we used a Cox proportional hazards model to determine whether high and low EC was associated with the incidence of LGI-GVHD. Interestingly, we found that EC 3.2.1.8 xylanase and EC 3.2.1.177 xylosidase gene abundances, but not those of EC 3.2.1.156 xylanase and EC 3.2.1.37 xylosidase, showed marked and significant differences in LGI-GVHD risk (Figure 4C-D; supplemental Figure 1). Overall, our data suggest an important role of carbohydrate degradation, specifically the degradation of xylose-containing polysaccharides, in mediating LGI-GVHD risk and that carbapenem exposure mediates this effect by reducing specific carbohydrate-degrading bacteria and functionalities in the microbiome.

Discussion

The microbiome plays an integral role in immune regulation and contributes to LGI-GVHD pathogenesis in patients who undergo allo-HSCT.¹⁵ To date, microbiome diversity has been recognized as an important indicator of clinical outcomes in allo-HSCT, including GVHD incidence and survival.^{15,19} A diverse microbiome helps maintain host immunity through various bacterial functions and metabolites. For example, specific metabolites such as SCFAs are known to modulate the intestinal immune system and contribute to immune tolerance.^{4,5} The administration of butyrate, one of the SCFAs, has been shown to reduce the severity of GVHD in a preclinical model.⁴ In addition to SCFAs,

monosaccharides produced from polysaccharide degradation by microbiota have also been found to play a significant role in mediating the disease activity of LGI-GVHD.² Therefore, understanding the function of beneficial microbes in addition to the microbiome composition is important to develop strategies to mitigate potential toxicities of allo-HSCT.

Here, we show that specific taxa including *B ovatus*, among other members of the *Bacteroides* and *Parabacteroides* spp., may play a beneficial role in preventing LGI-GVHD onset. We found that these bacteria correlated with multiple CAZyme gene abundances, consistent with previous reports that show their function as degraders of broad types of dietary polysaccharides.¹⁷ The role of *P merdae* is controversial in literature, with some demonstrating its presence in LGI-GVHD and contribution to ulcerative colitis (UC).²⁰ Others have shown *P merdae* may play a beneficial role in murine models of UC.²¹ *P distasonis* has been shown to potentially protect against models of colitis via promoting ILC3 in the gut and by reducing proinflammatory cytokines and shifts in microbial composition.^{22,23} Consistent with our and other authors' previous data, *B ovatus* has a beneficial effect in LGI-GVHD and other colitis models.^{2,24-26} In general, *B ovatus* does not show any mucus degradation function and although *P merdae* and *P distasonis* can degrade mucus, their function is reduced compared with traditional mucus degraders such *B thetaiotaomicron* and *B fragilis*.¹⁷ Therefore, the abundance of dietary fiber-degrading bacteria with minimal mucus-degrading function during allo-HSCT may predict the risk of LGI-GVHD.

Currently, the dietary polysaccharides that are key to suppress LGI-GVHD are unknown. To date, a few polysaccharides such as fructo-oligosaccharide,²⁷ galacto-oligosaccharide,²⁸ and resistant potato starch²⁹ have been investigated in the clinical setting of allo-HSCT and shown some potential to reduce GVHD. We recently found that xylose supplementation and xylose-containing polysaccharide degradation by *B ovatus* effectively suppress the severity of GVHD in a preclinical model.² In addition, other groups reported that xylan, one of the xylose-containing polysaccharides, has been shown to reduce the severity of colitis in UC mouse models.^{3,30,31} Based on these findings, we focused on the impact of xylanase and xylosidase, both of which contribute to xylose-containing polysaccharide degradation, on LGI-GVHD in this study and found the beneficial role of xylanase and xylosidase in mediating the risk of LGI-GVHD. Xylanase activity may mediate this benefit by increasing the SCFA, butyrate.³² Consistent with our data, β -xylosidase has been shown to be lower in patients with inflammatory bowel disease than in healthy volunteers.³³ Notably, xylanase gene abundance is affected by carbapenem exposure, suggesting that carbapenem exposure is associated with the risk of LGI-GVHD via reduced enzyme activity of xylanase in the intestine. Dietary interventions that promote the delivery of xylose-based polysaccharides or targeted and timed

Figure 3. Carbohydrate degradation differentiates those at risk for the development of LGI-GVHD, with contributions from specific taxa. (A) CAZyme α -diversity measured by the Shannon index stratified by LGI-GVHD status. (B) PCoA plot showing β -diversity measures stratified by LGI-GVHD status. *P* value was analyzed with PERMANOVA. (C) Pie chart showing the number and classification of differentially abundant CAZymes using DESeq2. *P* values were adjusted by the Benjamini-Hochberg correction. (D) Balloon plot showing enzyme function and families that favor non-LGI-GVHD. (E) Bar graph showing the number of statistically significant differentially abundant CAZymes grouped by family. Red and blue in the heat map on the left indicate substrate degradation and nonsubstrate degradation, respectively. (F) Network analysis using logit-transformed Pearson correlation of normalized CAZyme and species of interest. (G) The number of CAZyme families linked to specific bacterial species of interest. (H) Differential abundance analysis of PUL genes from bacterial species, with CAZyme annotation as labeled. *P* value was analyzed via DESeq2 with the Benjamini-Hochberg FDR correction.

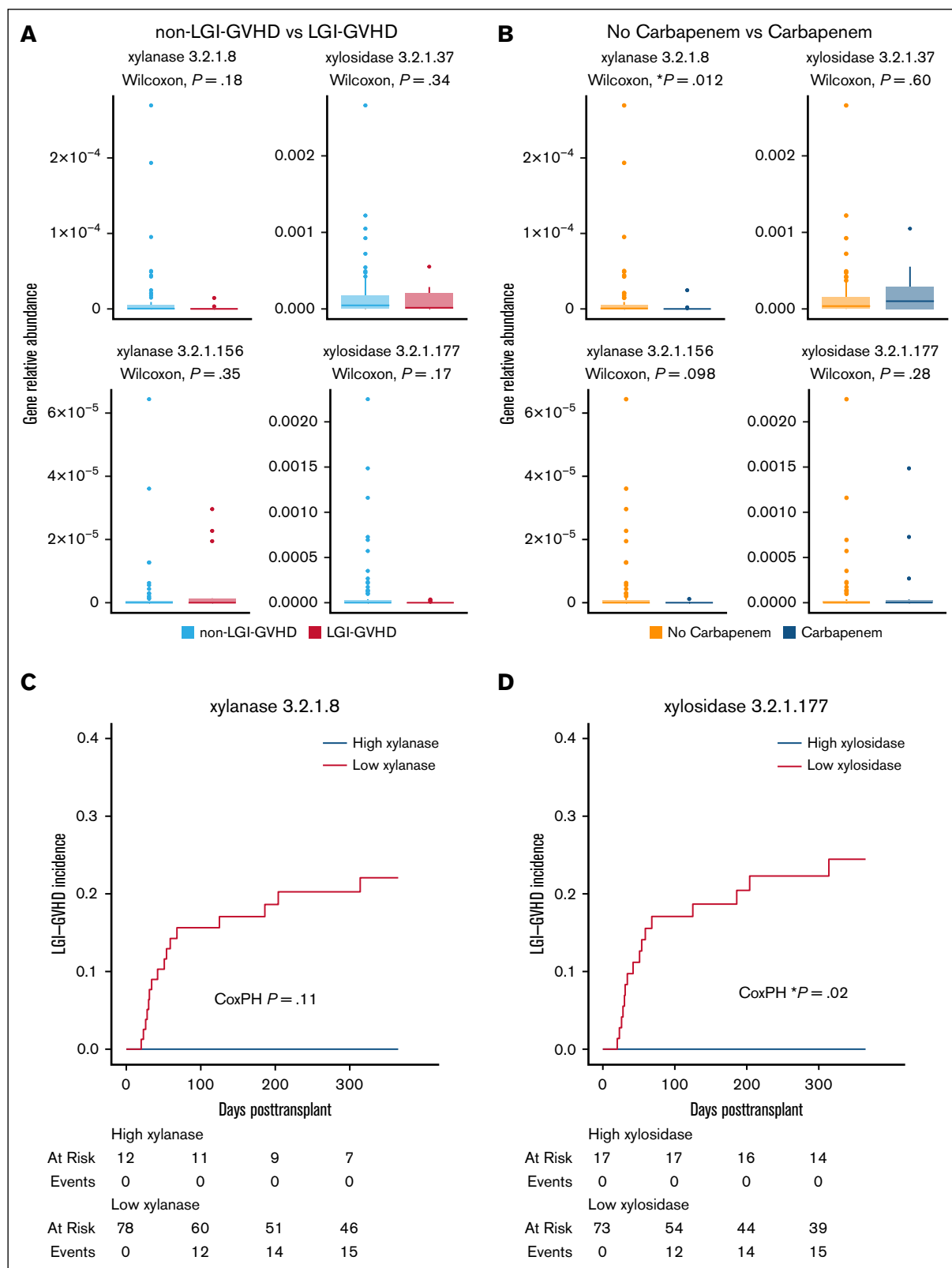


Figure 4. Xylanase and xylosidase activity differentiate those at risk for LGI-GVHD. (A) Comparison of the relative abundance of specific xylanase and xylosidase enzyme classes between patients with LGI-GVHD vs non-LGI-GVHD. (B) Comparison of the relative abundance of specific xylanase and xylosidase enzyme classes between carbapenem- and vs patients not exposed to carbapenems. (C) Kaplan-Meier curve showing the incidence of LGI-GVHD stratified by relative abundances of a specific xylanase enzyme class. (D) Kaplan-Meier curve showing the incidence of LGI-GVHD stratified by relative abundances of a specific xylosidase enzyme class. CoxPH, cox proportional hazards model.

release of xylose in the intestine may prove crucial to helping mitigate the risk of GVHD.

The retrospective nature and the small cohort of patients with LGI-GVHD compared with non-LGI-GVHD patients are limitations of our study. In addition, our data need to be validated in a separate independent cohort. There is heterogeneity in the timing of stool collection relative to LGI-GVHD onset, which may be a confounder. Our study does reinforce a prior study by Zeng et al, showing that there were no differences in α - and β -diversity measures in the fecal microbiome when comparing those who went on to develop LGI-GVHD and those who did not.³⁴ However, there have been 2 prior studies using 16S rRNA sequencing demonstrating an association between lower stool microbiome α -diversity and subsequent GVHD risk and transplant-related mortality.^{19,35} These discrepancies may partly be due to differences in sequencing techniques. Although there are no significant differences in allo-HSCT-related clinical characteristics between patients with non-LGI-GVHD and patients with LGI-GVHD except for acute GVHD grade, we were unable to tabulate dietary habits and caloric intake that may confound the microbiome composition and CAZyme of our results. Further studies are needed to investigate how to maintain beneficial microbial function, especially carbohydrate-degrading function, during allo-HSCT and whether the intake of xylose-containing polysaccharides is a key factor in preventing LGI-GVHD onset. In addition to the established risk factors for the development of GVHD such as HLA mismatching, older age of donor and recipient, conditioning and GVHD prophylaxis, multiple factors with contributions from diet and the microbiome likely contribute to LGI-GVHD risk.³⁶ These include the lack of dietary fiber consumption and carbapenem antibiotic use, which together can alter the composition and function of the microbiome, decreasing SCFA production and epithelial integrity and increasing mucolytic function that can increase the risk of LGI-GVHD.

In summary, specific beneficial taxa including *B ovatus*, *P distasonis*, and *P merdae* and xylanase and xylosidase activity are associated with the reduced risk of developing LGI-GVHD. The reduced abundances of these bacteria and xylanase and xylosidase activity by antibiotic exposure to carbapenems, including meropenem, may lead to an increased incidence of LGI-GVHD, supporting a potential mechanism by which carbapenem is associated with the risk of LGI-GVHD. This understanding of specific functional changes of microbes allow for specific prebiotic targets and interventions that may reduce the risk of clinical LGI-GVHD among patients who receive allo-HSCT.

Acknowledgments

The authors acknowledge the support of the High Performance Computing research facility at The University of Texas MD Anderson Cancer Center for providing computational resources that have contributed to the research results reported in this article. They would also thank the Fondren Inflammation Collaborative for their generous support and Rachael Whitehead for her assistance with helping create the visual abstract.

This work was supported by the National Institutes of Health (NIH)/National Heart, Lung, and Blood Institute (grant 2R01HL124112-06) (R.R.J.), Cancer Prevention & Research Institute of Texas (RR160089) (R.R.J.), NIH/National Cancer Institute Cancer Center Support (CORE) Grant (5P30CA016672-42)

(R.R.J.), NIH/National Heart, Lung, and Blood Institute (grant R01 HL158796) (C.B.P. and R.R.J.), NIH/National Cancer Institute Cancer Center Support Grant (grant P30CA016672; Biostatistics Resource Group) (C.B.P.), Andrew Sabin Family Fellowship (C.B.P.), American Society for Transplantation and Cellular Therapy New Investigator Award (E.H.), Amy Strelzer Manasevit Award from the National Marrow Donor Program Be the Match (E.H.), John Hansen Research Grant from DKMS Stiftung Leben Spenden (DKMS-SLS-JHRG-2023-02) (E.H.), Houston Methodist Clinical Scholars Award (C.F.), and NIH/National Institute of Allergy and Infectious Diseases (grants K24AI121296, R01AI134637, R01AI148342-01, and P01AI152999) (C.A.A.).

Authorship

Contribution: C.F., E.H., and R.R.J. conceptualized the research; C.F., E.H., T.H., and R.R.J. designed the research methodology; C.F., E.H., T.H., C.-C.C., I.K.G., I.I.F., L.K.M., M.R.O., C.A.S., R.K.E.-H., A.N.B., J.L.K., M.A.J., S.S.A., T.M.H., Y.J., W.-B.T., R.P., A.E., A.M., M.S., A.D., N.J.A., J.A.W., C.B.P., G.R., T.A.-J., A.M.A., J.J.M., R.E.C., E.J.S., E.M., C.A.A., and R.R.J. conducted the research investigation; C.F., E.H., and T.H. visualized the research; R.R.J. and E.H. acquired funding; C.F., R.R.J., and E.H. performed project administration; R.R.J. and E.H. provided supervision; C.F. wrote the original draft; and T.H., R.R.J., and E.H. contributed to review and editing of the manuscript.

Conflict-of-interest disclosure: R.R.J. has served as a consultant or advisory board member for Merck, Karius, MaaT Pharma, LIS Cure, Seres Therapeutics, Kaleido, Nestle, Probiotics plus, and Prolacta; and has received patent license fees or stock options from Seres Therapeutics and Kaleido. E.J.S. has served as a consultant or advisory board member for Adaptimmune, Axio, Navan, FibroBiologics, New York Blood Center, Zelluna Immunotherapy, Orca Biosystems, and Celaid Therapeutics; and has received patent license fees from Takeda, Prana X and Affimed. E.H., M.A.J., J.L.K., and R.R.J. are inventors on a patent application by The University of Texas MD Anderson Cancer Center supported by results of the current study titled, "Methods and Compositions for Treating Cancer therapy-induced Neutropenic Fever and/or GVHD."

ORCID profiles: C.F., 0000-0002-7117-6885; C.-C.C., 0000-0003-0050-1705; L.K.M., 0009-0005-1698-6657; J.L.K., 0009-0000-9360-4898; M.A.J., 0000-0003-3404-2535; S.S.A., 0009-0007-0520-1089; T.M.H., 0000-0001-8601-9949; Y.J., 0000-0001-7599-7042; W.-B.T., 0000-0001-8210-5882; R.P., 0000-0002-6875-0954; A.E., 0009-0005-2743-6722; A.M., 0009-0002-6241-9942; M.S., 0000-0002-8713-572X; A.D., 0009-0001-3158-2846; N.J.A., 0000-0002-3808-8576; C.B.P., 0000-0003-3316-0468; T.A.-J., 0009-0003-7220-2314; A.M.A., 0000-0002-2498-8573; J.J.M., 0000-0003-1196-1715; R.E.C., 0000-0002-4314-5037; E.J.S., 0000-0003-3947-4126; R.R.J., 0000-0002-5434-439X; E.H., 0000-0002-4824-8844.

Correspondence: Robert R. Jenq, Department of Hematology and Hematopoietic Cell Transplantation, City of Hope National Medical Center, 1500 East Duarte Road, Duarte, CA 91010; email: rjenq@coh.org; and Eiko Hayase, Department of Medicine, Baylor College of Medicine, 1 Baylor Plaza, ABBR, R531G, Houston, TX 77030; email: eiko.hayase@bcm.edu.

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